LKT augmented GRU with a 128-unit prediction head

Saved model: augmented__head128__seed1 | restored epoch 24 | 31,585 parameter entries

A Inputs available before response t

5 numeric coordinates (in saved order)
log(1 + student time gap in seconds)
intercept; student-item logitdec;
student-item PPES; student logitdec
Standardize using training-set statistics

Current item ID: KC..Default.
144 known items

Learned item embedding: 16 values
Table: 145 rows × 16 columns
144 known rows + 1 reserved zero row

Current skill ID: KC..Cluster.
36 known skills

Learned skill embedding: 16 values
Table: 37 rows × 16 columns
36 known rows + 1 reserved zero row

Concatenate: 5 numeric + 16 item + 16 skill = 37 coordinates
Input dropout 0.1 produces x(t), shared by prediction and memory update

Embeddings are trained jointly with the network. The centered intercept coordinate is zero; no problem-text embedding is used.

B Predict the current response

Previous top-layer memory

Previous h3(t−1): 32
+ current x(t): 37
69 coordinates

Linear 69 → 128 (with bias)
ReLU
Head dropout 0.1

Linear 128 → 1
(with bias)
One logit

Sigmoid + clipping
p(correct at trial t)
ε = 10⁻⁵

Current response y(t) does not enter this prediction. No separate inference network or Gaussian latent is present.

C Observe the answer, then update all three GRU layers

Observed response y(t)
0 = incorrect; 1 = correct

x(t): 37

Concatenate [x(t), y(t)]
37 + 1 = 38 coordinates

h1(t−1)
32 previous values

GRU layer 1
32 units
Own state

h1(t): 32

Dropout
0.1

h2(t−1)
32 previous values

GRU layer 2
32 units
Own state

h2(t): 32

Dropout
0.1

h3(t−1)
32 previous values

GRU layer 3
32 units
Own state

h3(t)
32 values

Retain all three states for the next trial: [h1(t), h2(t), h3(t)] = 3 × 32 values

Each new participant starts with zero states. Updated memory can affect the NEXT prediction, p(t+1).

Dropout is active only during training. Binary cross-entropy trains the logits; all learned weights stay fixed during continuation.

D Inside each GRU layer: exact Torch gate equations

Layer input u(t) has 38 coordinates in layer 1 and 32 in layers 2 and 3. Every layer has its own weights and biases.

Layer input
u(t)

Previous state
h(t−1): 32

Separate learned affine maps
for g in {reset r, update z, new n}

a_g = W_ig u(t) + b_ig

c_g = W_hg h(t−1) + b_hg

Each a_g and c_g has 32 values

UPDATE gate
z(t) = sigmoid(a_z + c_z)

RESET gate
r(t) = sigmoid(a_r + c_r)

CANDIDATE state
n(t) = tanh(a_n + r(t) ⊙ c_n)

z(t)

r(t)

n(t)

Mix candidate and old state

h(t) = (1−z(t)) ⊙ n(t)
+ z(t) ⊙ h(t−1)

New state
h(t): 32

Old state: identity carry path

⊙ means elementwise multiplication. The reset gate multiplies the full recurrent candidate projection, including its bias.

All embedding, GRU and head weights are learned jointly through backpropagation. The diagram shows their forward use.

The head consumes the previous top-layer state; it never consumes the current answer or the newly updated state.

Architecture from the original 2.0.0.9009 checkpoint; dimensions, equations and real-history replay verified using maintained LKT 2.0.0.9010.

Source: R/lkt-model-gru.R and R/lkt-gru-prediction.R. Original fitted weights and historical results were not modified.